

748

teleoperated episodes collected across three sites

508

gradable on-hardware autonomous rollouts

3 sites

Pool (Control) · *Coralita X127* Wreck · Manoel Island Marina, Malta

Two Task Families

Grab the Red Rod · Moor onto the Rope/Ladder/Post

47.9% vs 7.9%

Pooled success — ACT vs. SmolVLA respectively, across all tasks/sites

Introduction

- Marine archaeologists routinely monitor sites and retrieve specimens under poor visibility, dynamic currents, and the physiological limits of human divers. Automating these operations could accelerate scientific research while reducing diver risk.
- BC-based policies, particularly Vision-Language-Action (VLA) models, have shown remarkable generalization to novel tasks and objects, making them an appealing candidate for underwater manipulation.
- However, they are trained almost exclusively on terrestrial datasets featuring stable lighting and fixed-base manipulators, while underwater vehicles operating in marine environments experience disturbances.

Do these policies retain their generalization capabilities in the wild, under marine field conditions, and across diverse archaeological sites?

Contributions

- A real-world case study** of BC manipulation models deployed on an underwater Remotely Operated Vehicle (ROV) at a Mediterranean shipwreck for archaeology and conservation tasks.
- An empirical evaluation** of BC model performance under varying lighting, visibility, and environmental disturbances.
- Analysis of failure modes** when transferring BC models underwater, and key directions for the development of robust marine embodied intelligence.

Conclusion

1 · A vision backbone mismatched to the domain

A VLA's value is a backbone pretrained on terrestrial data. If frozen, every underwater frame is out of distribution. Fine-tuned on our small, noisy dataset, SmolVLA overfits and loses generalization. ACT's small encoder, alternatively, trains end-to-end on in-domain demonstrations.

2 · Persistent disturbance under a slow policy

In marine field environments, the ROV is jostled in all six DOF, so movements are prone to the accumulation of error. SmolVLA, a heavier model, replans infrequently, so a chunk conditioned on a stale observation drifts before the next inference step. ACT's lighter architecture replans faster, staying closer to the loop.



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lair-research-nyu.github.io/Embodied-at-Sea

Methodology

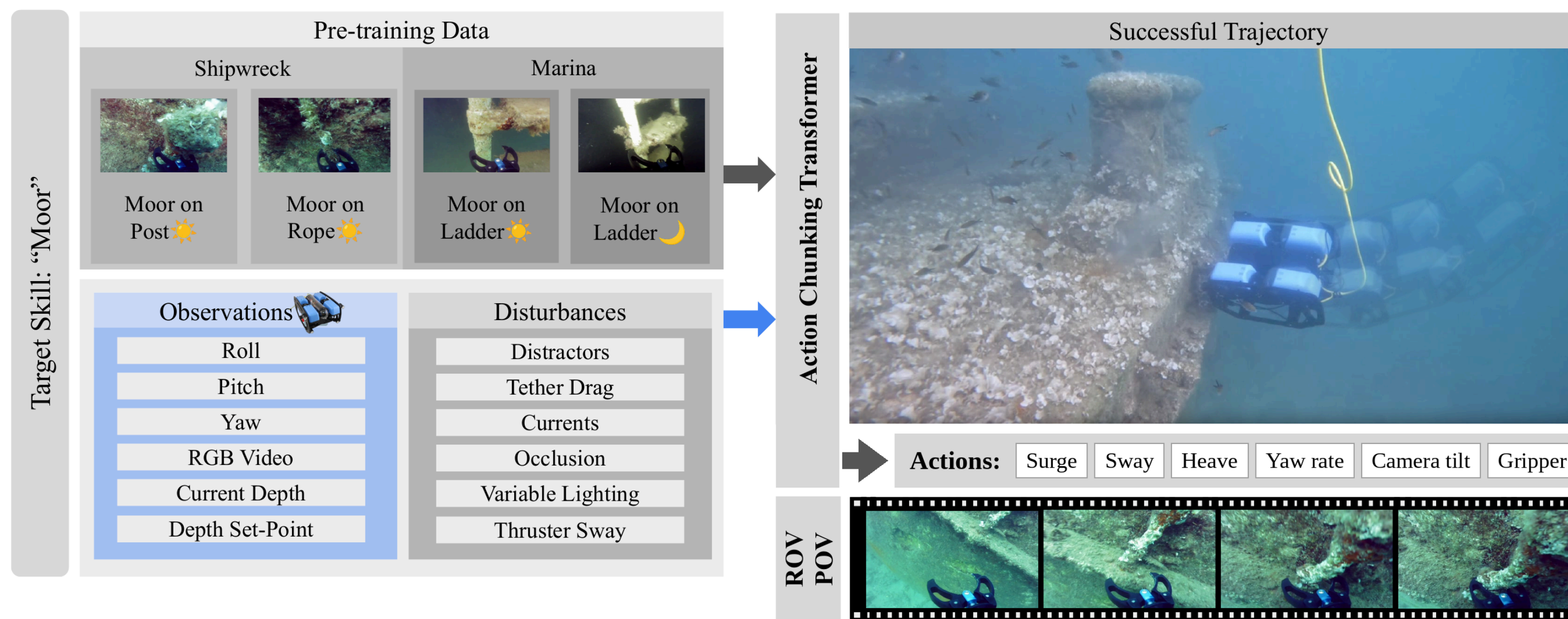


Fig. 1. Overview of Autonomous Underwater Mooring using Action Chunking Transformers (ACT). (Left) Pre-training data collected across shipwreck and marina environments incorporating multi-modal observations and real-world disturbances. These include marine life distractors, tether drag, ocean currents, visual occlusion from water particulate, variable lighting based on depth or time of day, and thruster sway inherent to fluid dynamics. (Right) Autonomous policy execution showing closed-loop mooring trajectory on the *Coralita X127* shipwreck.

Tasks and sites

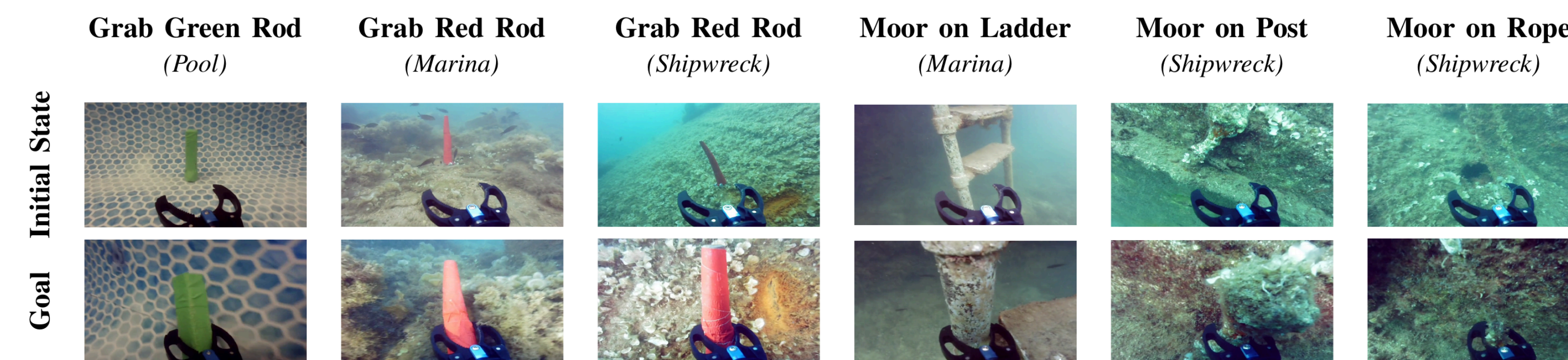


Fig. 2. Initial state and goal for each task variant. The pool was fully lit with no currents; the Marina and the *Coralita X127* were flown day and night under varying disturbance.

Results

Overall Success Rates for ACT and SmolVLA by Task and Site

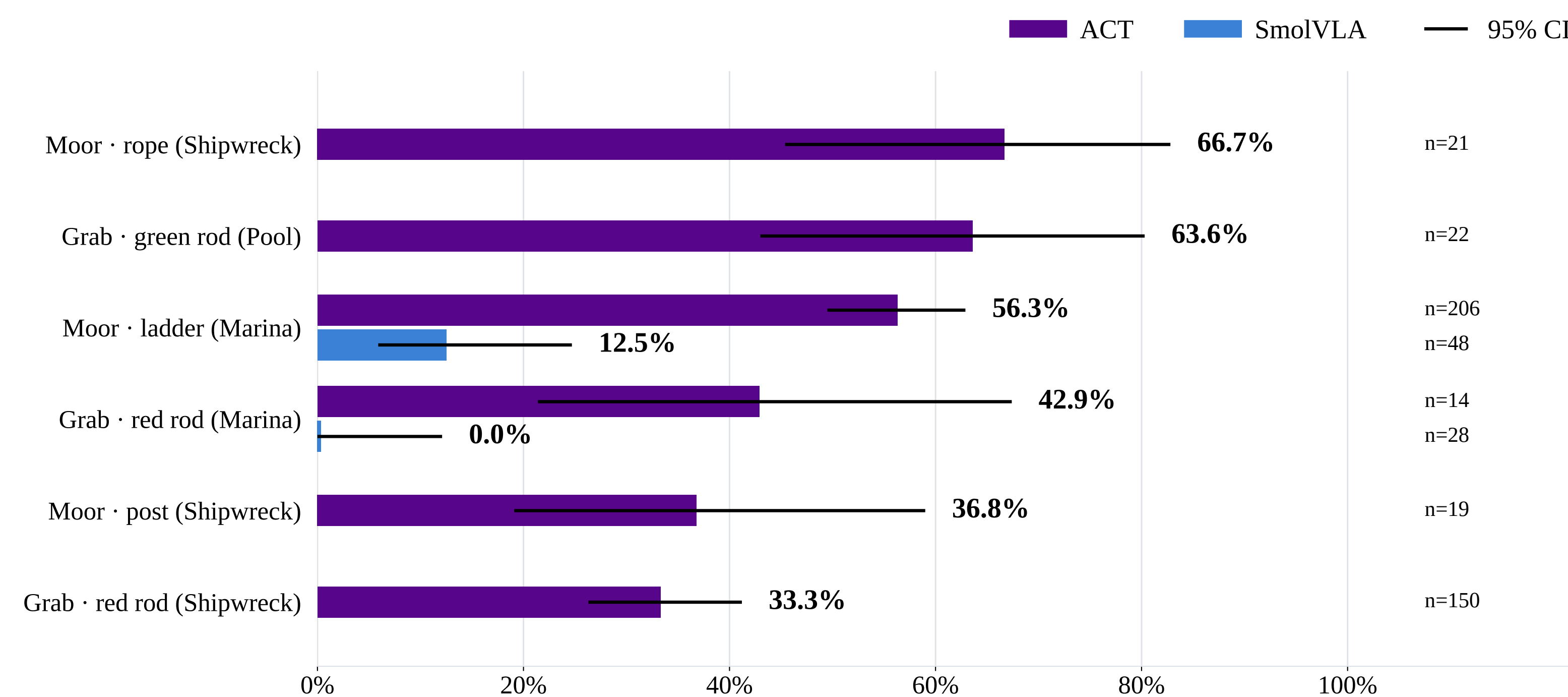


Fig. 3. Success rate by task and location. ACT consistently outperforms SmolVLA, and Mooring is the best-performing task family for both architectures.

Results (cont.)

Daylight vs. Night Time Deployment Success Rates for ACT

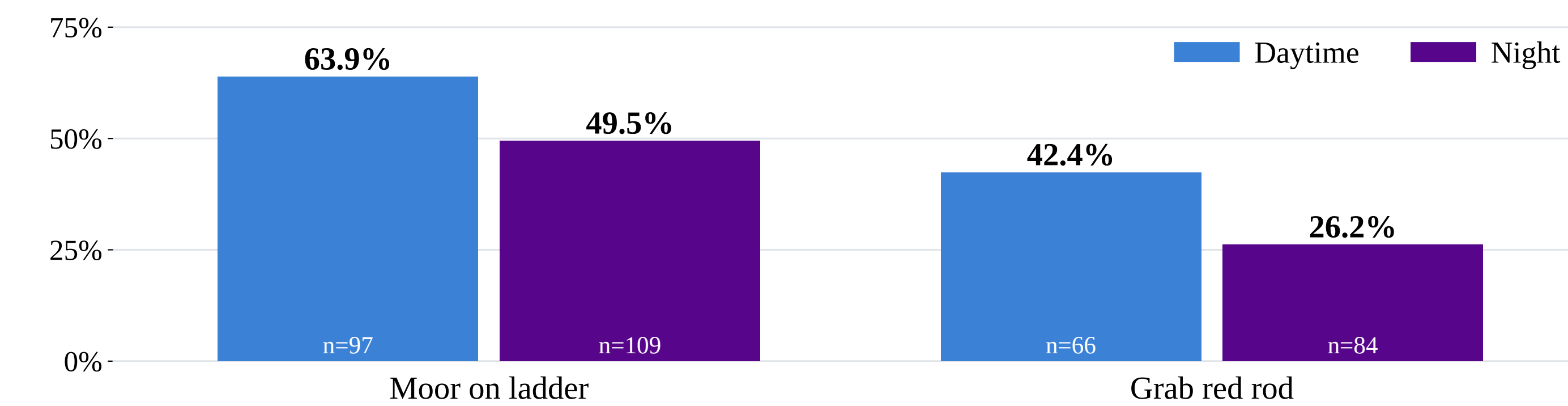


Fig. 4. ACT only: moor on ladder at Manoel Island Marina (n=206) and grab red rod on the *Coralita X127* (n=150), split by time of day. Both tasks drop in success at night. Occluded vision and turbulent waters are common in Mediterranean waters during the night.

Emerging behaviors

Behavior #1 Diving to the target

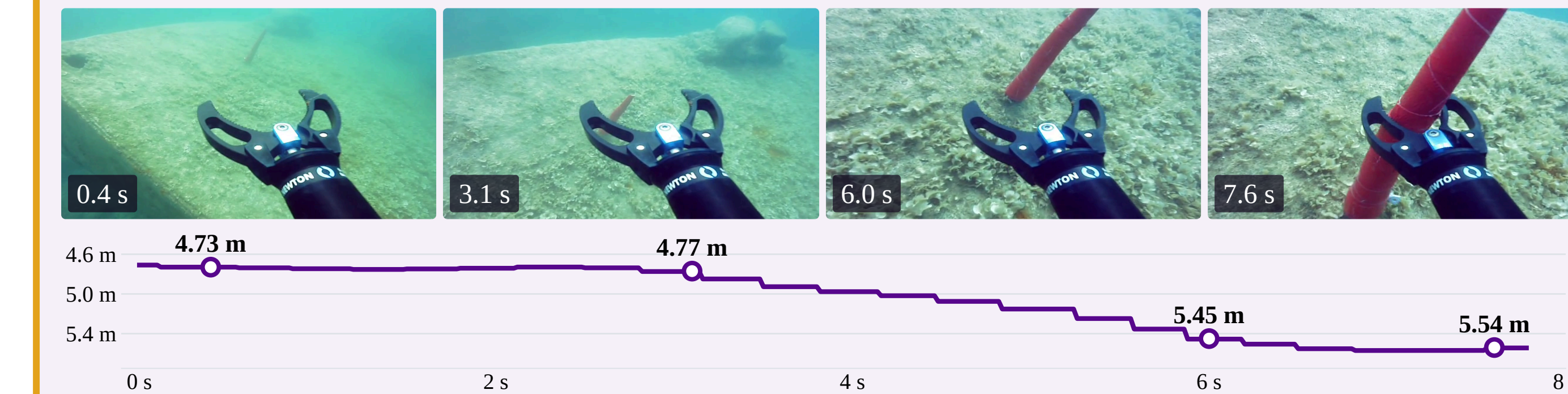
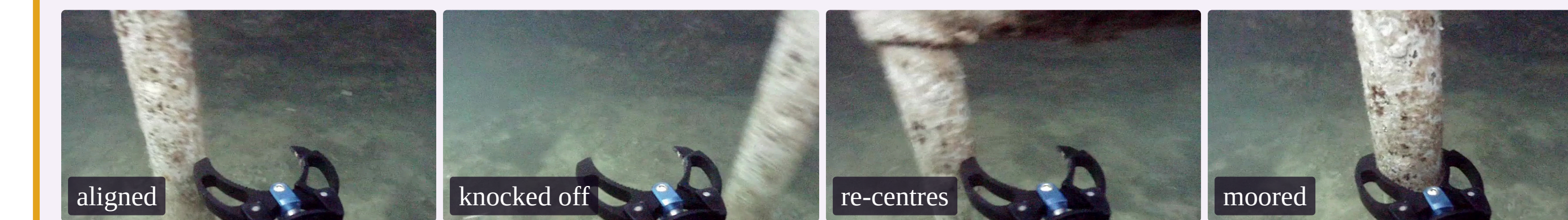


Fig. 5. *Grab red rod* on the *Coralita X127*: the rod sits below the robot, so the policy dives more than 0.8 meters in under 8 seconds to achieve success. Depth (see Fig. 1), known in standard marine-vehicle convention as “heave,” is included in the action space. On the *Coralita X127*, tasks were done at varying depths, making heave crucial for success.

Behavior #2 Recovering from failure during night time turbulence

Moor on ladder — pushed off the pole, re-acquires it



Grab red rod — misses, backs off, re-approaches



Fig. 6. ACT showed strong recovery trajectories under rough night-time currents at Manoel Island Marina; SmolVLA showed none. Robustness to turbulence is necessary for successful marine manipulation.

Future Works

- Extending the observation space** with IMU data would allow BC policies to account for ROV motion induced by hydrodynamic disturbances, reframing underwater manipulation as a partially dynamic task rather than a static one.
- Improving **chunked inference speed**, as explored in DynamicVLA, would allow heavier VLAs to replan quickly enough to track disturbances in near real-time.
- Finally, collecting **additional trials** across *Coralita X127* and other wrecks, evaluating longer-horizon tasks such as tether untangling, and progressively extending deployments to greater depths would allow policies to advance toward robust underwater embodied intelligence.